Condominium Property Management and Security System

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BUSINESS SCENARIO

A condominium community has multiple residential units. The property administrator manages tenant information and monthly dues, while security personnel monitor residents, visitors, and service personnel entering the property.

PROBLEM

- Manual management can cause:
- Difficulty organizing tenant records
- Slow resident verification
- Unauthorized visitor access
- Errors when calculating monthly dues
- Errors in calculating late penalties
- Difficulty tracking security access records

SOLUTION

The system provides one centralized console application that allows users to:

Register Tenant → Verify Resident → Approve/Deny Visitor → Record Access → Calculate Dues → Calculate Penalties

The system specifically checks the visitor's target unit and host approval before granting visitor access.

CONDOMINIUM ACCESS & DUES SYSTEM LOGIC & ARCHITECTURE

Data Flow:

Tenant Directory Management

- Register
- View
- Search
- Update
- Remove

Calculate Dues & Penalties

- Fetch Unit Info
- Calculate Association Dues
- Calculate Tiered Penalty
- Print Statement

Gate Access Control

- Process Entry
- View Resident Log
- View Visitor Log

Variable Scope

Scope	Type	Example	Purpose
Static Global	Collections & Constants	tenantList, ASSOCIATION_DUES_PER_SQM	Shared across system lifetime
Instance (Tenant Object)	Primitive & Object Ref	unitNumber, name, baseRent	Encapsulated specific record data
Local Method	Function Input/ Calculations	daysOverdue, penaltyFee, visitorName	Specific to module execution

Decision Condition

Condition	Condition
Is Unit Registered?	Yes → Fetch Record / Process Entry No → Input Manually / Deny Entry
Days Overdue?	Tier 1 (0-5d): Flat Penalty Tier 2 (>5d): Tiered Daily Penalty
Host Resident Approved?	Yes → Granted (Visitor Pass) No → Denied

LIVE CONSOLE DEMONSTRATION

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|   ADMIN / PROPERTY ADMINISTRATOR DASHBOARD |
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[1] Tenant Directory Management
[2] Calculate Monthly Dues & Late Penalties
[3] Security Gate Access Log
[4] Exit System

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Enter Choice: |
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|        SECURITY / STAFF DASHBOARD        |
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[1] Process Entry Request
[2] View Resident Entry Logs
[3] View Visitor Access Logs
[4] Back to Main Menu

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Enter Choice: |
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